Techniques for Selecting Features in Medical Data

Mahade Hasan, Farhana Yasmin, and Yu Xue

Abstract

In machine learning and medical data analysis, feature selection is crucial because it helps find the most important characteristics while removing those that are unnecessary or redundant. Many features in datasets used in medical applications may add noise or complex computations, and many of these features do not substantially improve predictions. This chapter explores various feature selection techniques, broadly categorized into supervised and unsupervised methods. Supervised methods for feature selection rely on annotated data to determine the importance of features. These include filter, wrapper, and embedded methods. Filter methods evaluate the significance of each feature separately using statistical tests such as Chi-square, ANOVA, and Mutual Information. These methods are computationally efficient and scalable, but do not consider feature interactions. Wrapper methods, such as Recursive Feature Elimination (RFE) and Sequential Feature Selection, evaluate feature subsets by training a predictive model, leading to improved accuracy but higher computational costs. Embedded methods, including regularization techniques like Lasso and Ridge regression, perform feature selection as part of the model training process, balancing accuracy and efficiency. In contrast, unsupervised feature selection methods do not require labeled data and rely on data dimension criteria to reduce the number of features. These methods are particularly useful in exploratory data analysis and clustering applications. By utilizing suitable feature selection techniques, researchers can improve model performance, enhance interpretability, and reduce overfitting. This chapter explores the benefits and drawbacks of each method, with a focus on their applications in medical research, including genomic analysis, medical imaging, and electronic health records.

Keywords

Embedded method · Feature selection · Filter method · Medical data analysis · Wrapper method

1 Introduction

Medical data is inherently complex, often characterized by high dimensionality, noise, and redundant information. In order to simplify this data, feature selection is essential because it finds the most relevant factors that have a major impact on the predictive modeling process. By reducing the chance of overfitting, it enhances predictive model performance and improves generalization (Salsabila et al. 2025; Akter et al. 2025). Additionally, feature selection enhances interpretability, allowing clinicians to focus on key variables, which improves the trust and usability of machine learning models. Reducing the number of features minimizes computational complexity, decreases memory usage, and speeds up training and inference times. Furthermore, by identifying the most important features, researchers can gain insights into disease mechanisms, treatment effectiveness, and potential biomarkers. In many cases, medical datasets (Khan et al. 2024) have a limited number of patient records. Feature selection mitigates the curse of dimensionality, ensuring models can still perform well despite small sample sizes. Medical data presents several challenges that make feature selection a necessary step before applying machine learning techniques

M. Hasan (✉) · Y. Xue
School of Software, Nanjing University of Information Science and Technology, Nanjing, China
e-mail: mhasan@nuist.edu.cn

Y. Xue
e-mail: xueyu@nuist.edu.cn

F. Yasmin
School of Computer Science, Nanjing University of Information Science and Technology, Nanjing, China
e-mail: farhana@nuist.edu.cn

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

A. Nag et al. (eds.), *Feature Fusion for Next-Generation AI*, Sustainable Artificial Intelligence-Powered Applications,
https://doi.org/10.1007/978-3-031-94386-7_3

(Yasmin et al. 2023a). High dimensionality is a common issue, especially in medical imaging, genomics, and electronic health records. A single microarray gene expression dataset may contain tens of thousands of genes as potential features. Medical imaging data, such as MRI or CT scans, can generate millions of pixel values, while patient records may include thousands of structured and unstructured features, such as lab results, clinical notes, and demographics. High-dimensional data increases the risk of overfitting, making it crucial to identify the most informative features. Another major challenge in medical data is noise, which can stem from various sources. Measurement errors, variability in lab test results, and sensor inaccuracies contribute to data inconsistencies. Human errors, such as mistakes in data entry or diagnosis, can introduce bias, while missing values due to incomplete documentation further complicate model training. Noisy features (Ergul Aydin and Kamisli Ozturk 2024) can mislead models, resulting in unreliable predictions. Feature selection helps filter out irrelevant and misleading variables, improving model robustness and accuracy. Redundancy is another issue in medical datasets (Yasmin et al. 2023b), as many features are highly correlated. Multiple variables may provide the same information, leading to inefficiency. For instance, blood pressure measurements taken at different times may show strong correlations, multiple gene expressions may be co-regulated, and MRI pixel intensities in adjacent regions may convey similar information. Removing redundant features improves model efficiency and avoids unnecessary complexity, ensuring that models focus only on the most critical attributes.

Feature selection directly impacts the precision and practicality of medical ML models. Reducing the number of features minimizes the risk of overfitting, ensuring that predictions remain accurate when tested on new, unseen data. By focusing on a smaller set of critical features, machine learning models become more interpretable, making it easier for clinicians to understand the reasoning behind predictions. For example, in heart disease prediction, selecting features like cholesterol level and blood pressure makes the model more interpretable than using hundreds of physiological parameters. Similarly, in cancer diagnosis, highlighting a few key genes associated with tumor growth can guide further research and treatment plans. Feature selection aligns with the goals of Explainable AI (Hossain et al. 2025) (XAI) by ensuring that machine learning models provide insights that can be understood and trusted by medical professionals, a crucial aspect for regulatory approval and adoption in clinical settings. Feature selection is widely used in various medical applications, including disease diagnosis, prognosis prediction, and personalized treatment planning. In medical diagnosis, selecting relevant features helps in detecting diseases more accurately. Examples include heart disease prediction using major risk factors, including age, blood pressure, cholesterol levels, and ECG readings, cancer detection by identifying critical biomarkers in genomic data, and diabetes detection through features like glucose level, insulin resistance, and BMI. In prognosis prediction, feature selection aids in predicting disease progression and patient outcomes. For example, in survival prediction for cancer patients, genetic markers and clinical parameters help estimate patient survival rates, while early indicators from brain imaging and cognitive test results assist in predicting the progression of Alzheimer's disease. By choosing the most relevant patient-specific features, machine learning models can recommend optimal treatments. In drug response prediction, genetic markers that influence how patients respond to specific medications are selected (Rahman et al. 2024). In radiotherapy planning, the most relevant imaging features are chosen to optimize radiation doses for cancer patients. Personalized diabetes management benefits from identifying lifestyle and biological factors that impact glucose control. Medical imaging analysis also relies on feature selection to extract the most informative features from complex scans such as MRI, CT, and X-rays. Brain tumor segmentation, for example, involves selecting relevant MRI features for tumor detection, while lung disease classification requires identifying key CT scan features to differentiate between pneumonia, tuberculosis, and COVID-19.

Feature selection is a crucial step in medical data analysis, addressing challenges like high dimensionality, noise, and redundancy. By selecting the most relevant features, we improve model accuracy, interpretability, and computational efficiency, making machine-learning models more practical for real-world medical applications. The next chapters will explore specific feature selection techniques and their applications in greater depth.

2 Fundamentals of Feature Selection

2.1 Definition Feature Selection

Feature selection refers to the process of choosing a subset of features from the original set of features that are most relevant to the predictive modeling task. It is a form of dimensionality reduction that retains the original features rather than transforming them, as in feature extraction methods like Principal Component Analysis (PCA) (Kumar Sahu and Fatma 2025). Mathematically, given a dataset D with n samples and p features, feature selection aims to find a subset $S \subseteq [1p]$ such that the performance of model M is optimized. The subset S typically contains k features, where $k \ll p$.

2.2 Objective of Feature Selection

A key step in machine learning is feature selection, which involves selecting a subset of relevant characteristics (variables, predictors) to be used in building a model. We go into further detail about the main advantages of feature selection below, using mathematical expressions where appropriate.

(a) ***Improving Model Performance***

- ***Reducing Overfitting***: When a model learns the noise or insignificant patterns in the training data, it is considered to be overfitting and will not generalize well to new data. Feature selection helps mitigate this by removing irrelevant or redundant features.
  - Let $X \in \mathbb{R}^{n \times p}$ be the feature matrix with n samples and p features.
  - Let $y \in \mathbb{R}^n$ be the target variable.
  - Let $S \subset \{1, 2, \ldots, p\}$ be the subset of selected features, where $|S| = k$ and $k \ll p$.
  - By reducing the feature space to k relevant features, the model complexity decreases, reducing the risk of overfitting. The reduced feature matrix is $X_s \in \mathbb{R}^{n \times k}$.
- ***Enhancing Generalization***: A model trained on a smaller, relevant feature subset generalizes better to unseen data. This is because the model focuses on the most informative features, avoiding noise. The generalization error $\in$ can be approximated as: $\in = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$.
  By reducing the number of features, the variance term decreases, resulting in better generalization.

(b) ***Reducing Computational Complexity***

- **Faster Training**: The computational cost of training a model is often a function of the number of features p. Reducing p to k significantly reduces the training time. For example, in linear regression, the time complexity of solving the normal equations $\hat{\beta} = \left(\left(X^T X\right)\right)^{-1} X^T y$ is $O\left(p^3\right)$. Reducing p to k reduces the complexity to $O\left(p^3\right)$.
- **Efficient Storage**: Storing fewer features reduces memory requirements. The storage complexity of the feature matrix X is reduced from $O(n \times p)$ to $O(n \times k)$.

(c) ***Enhancing Interpretability***

- **Simpler Models**: It is easier to analyze and explain models with fewer characteristics, which is important in medical applications where comprehension of the decision-making process is critical.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + \in, \qquad (1)$$

  where $k \ll p$ is the contribution of each feature x_i to the target y that is more transparent.
- **Domain Understanding**: By identifying the most crucial characteristics, feature selection offers insights into the underlying mechanisms. For example, in medical applications, selected features might correspond to specific biomarkers or clinical indicators.

(d) ***Handling High-Dimensional Data***

- **Curse of Dimensionality**: In high-dimensional spaces, data becomes sparse, and distances between points become less meaningful. This degrades the performance of many machine learning algorithms. The volume of a hypercube in high dimensions is concentrated in its corners, making distance metrics less effective. Feature selection reduces the dimensionality, mitigating this issue. The curse of dimensionality can be quantified by the ratio of the variance to the squared expectation of the distance between points:

$$\frac{\text{Var}(d)}{E[d]^2} \to 1 \quad \text{as } p \to \infty, \qquad (2)$$

  where d is the distance between points. Feature selection reduces p, improving the effectiveness of distance-based algorithms.

(e) ***Improving Data Quality***

- **Noise Reduction**: Irrelevant or noisy features can degrade model performance. Removing such features improves the signal-to-noise ratio (Singh et al. 2024). Let $X = X_{\text{signal}} + X_{\text{noice}}$, where X_{signal} represents noisy features. Feature selection removes X_{noice}, improving the quality of X_{signal}.
- **Handling Multicollinearity**: Unstable model estimates arise from multicollinearity, which happens

when characteristics are highly linked. Feature selection can address this by retaining only one feature from a group of correlated features (Rabiee et al. 2024). The correlation matrix R of the features can be used to identify multicollinearity. If $R_{ij} \approx 1$ for $i \neq j$, one of the features x_i or x_j can be removed.

3 Types of Feature Selection Methods

By identifying the most important characteristics and removing those that are redundant or unnecessary, feature selection techniques let machine learning models operate at their best with less complexity.

The hierarchical structure of feature selection techniques is depicted in Fig. 1, which divides them into supervised and unsupervised approaches. The three primary methods of supervised feature selection are the Filter Method, Wrapper Method, and Embedded Method. Unsupervised feature selection, on the other hand, eliminates less useful features using data dimension criteria, such as variance thresholds, rather than labeled data.

3.1 Filter Methods

Filter methods evaluate feature importance using statistical techniques before model training. These methods are independent of specific machine learning algorithms, making them computationally efficient and suitable for high-dimensional datasets (Natarajan et al. 2025; Rahmat et al. 2024) (Fig. 2).

The relationship between a feature A and the goal variable B is measured by mutual information, which is defined as

$$I(A, B) = \sum_{a \in A} \sum_{b \in B} P(a, b) \log \frac{P(a, b)}{P(a)P(b)}, \tag{3}$$

where A and B's joint probability distribution is denoted by $P(a, b)$, and their marginal distributions are denoted by $P(a)$ and $P(b)$. Another option for categorical features is the chi-square test, which measures the degree of independence

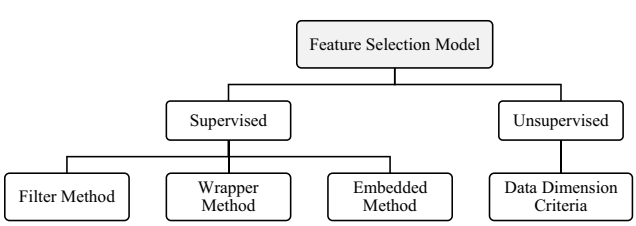

Fig. 1 Types of feature selection methods, such as supervised and unsupervised methods

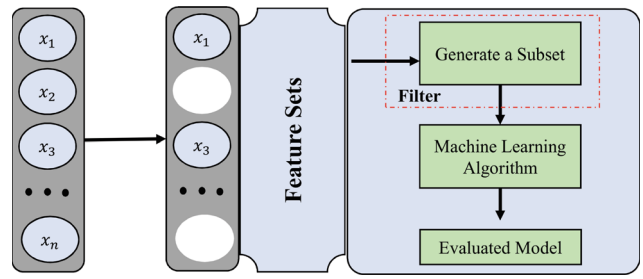

Fig. 2 Illustration of the filter-based feature selection method

between feature A and target B using

$$x^2 = \sum \frac{(O - Q)^2}{Q}, \tag{4}$$

where O represents observed frequency and Q represents expected frequency. Pearson correlation assesses the linear relationship between a feature A and the target B using

$$r = \frac{\sum (A_i - \overline{A})(B_i - \overline{B})}{\sqrt{\sum (A_i - \overline{A})^2 \sum (B_i - \overline{B})^2}}. \tag{5}$$

Features with high correlation with the target but low correlation among themselves are preferred to reduce redundancy.

3.2 Wrapper Methods

Wrapper methods assess feature subsets by iteratively training a model, capturing interactions between features but requiring higher computational resources. Recursive Feature Elimination (RFE) (Wang et al. 2025) eliminates the least important feature in each iteration based on model coefficients $\beta = \beta_1, \beta_2, \ldots, \beta_n$, where the feature with the smallest absolute $|\beta_i|$ is eliminated. Forward selection adds features incrementally if they enhance model performance, while backward elimination starts with all features and removes the least significant ones based on p-values. Genetic algorithms explore feature subsets using evolutionary principles like selection, crossover, and mutation to optimize feature selection (Song et al. Jan. 2025) (Fig. 3).

3.3 Embedded Methods

Embedded methods integrate feature selection during model training, balancing computational efficiency and feature interaction capture (Nemati et al. 2024). LASSO regression (Pak et al. 2025) applies L_1 regularization, forcing some feature coefficients to zero:

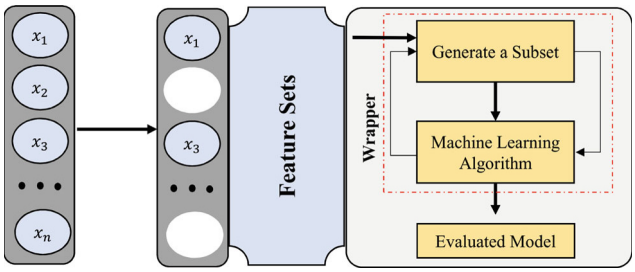

Fig. 3 Illustration of the wrapper-based feature selection method

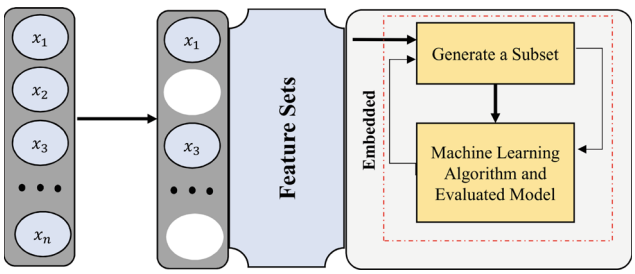

Fig. 4 Illustration of the embedded-based feature selection method

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} X_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|, \tag{6}$$

where λ controls coefficient shrinkage. Decision trees and ensemble methods, such as Random Forests, assign importance scores based on feature contributions to reducing impurity, often measured using Gini impurity or information gain. In Random Forests, feature importance is computed as

$$I(X_j) = \frac{1}{T} \sum_{t=1}^{T} I_t(X_j), \tag{7}$$

where $I_t(X_j)$ represents the importance of the feature X_j in tree t, and T is the number of trees. Features with higher scores are considered more informative (Fig. 4).

4 Evaluation Metrics for Feature Selection

Evaluating the performance of feature selection techniques is crucial to ensure that the chosen features improve model performance and generalize well to new data mentioned in Table 1. Common evaluation metrics include accuracy, Sensitivity (Recall), Specificity, AUC-ROC, F1-score, and Stability.

Table 1 Performance evaluation metrics[a]

Metrics	Metrics formulas	Evaluation focus
Accuracy (Yasmin et al. 2023b)	Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$	Finds the percentage of cases that are correctly classified
Sensitivity (recall) (Hema et al. 2024)	Sensitivity $= \frac{TP}{TP+FN}$	Calculates the percentage of true positives that were accurately detected
Specificity (Karampour et al. 2024)	Specificity $= \frac{TN}{TN+FP}$	Represents the percentage of real negatives that were accurately detected
F1-Score (Hasan et al. 2023)	F1 -Score $= \frac{TP}{TP+FP}$	Balances precision and recall

[a] TP stands for true positive, TN for true negative, FP for false positive, and FN for false negative

5 Data Preprocessing for Feature Selection

Effective data preprocessing is essential for ensuring that feature selection techniques yield meaningful and reliable results, particularly in medical datasets, which often contain missing values, imbalanced class distributions, and noise. Proper handling of these issues enhances model performance, interpretability, and generalizability.

5.1 Handling Missing Data in Medical Datasets

Handling missing data is a crucial step in preparing medical datasets, as missing values can introduce biases and reduce model reliability. Various imputation techniques address this issue, including mean, median, and mode imputation, which replace missing values with imputed values and statistical measures of the existing data. More advanced approaches, such as k-nearest neighbors (KNN) imputation, matrix factorization, and deep learning-based imputation, leverage complex relationships within the dataset to make accurate estimations. Multiple imputation methods generate multiple plausible values for missing entries and aggregate them to reduce uncertainty. In cases where messiness follows a pattern, domain-specific techniques incorporating expert knowledge can further improve data completeness.

5.2 Normalization and Standardization of Medical Data

Normalization and standardization are essential when working with medical datasets containing variables measured

on different scales. Normalization rescales feature values to a consistent scale fixed range, typically [0, 1] or [−1, 1], using min–max scaling:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}, \tag{8}$$

where $X_{\min}$, $X_{\max}$ represent the minimum and maximum values of the feature data. Standardization, on the other hand, transforms features to have zero mean and unit variance:

$$X_{\text{std}} = \frac{X - \mu}{\sigma}, \tag{9}$$

where μ is the mean and σ is the standard deviation of the feature data. Standardization is particularly useful when features have different units or scales, ensuring that each variable contributes equally to the model.

5.3 Dealing with Imbalanced Data in Medical Applications

Dealing with Imbalanced Data is crucial in medical applications, where minority class instances (e.g., positive disease cases) are often much rarer than negative cases. Resampling techniques help mitigate this issue, with oversampling methods such as SMOTE generating synthetic samples for the minority class, while undersampling reduces the number of majority class samples. Cost-sensitive learning assigns higher misclassification penalties to the minority class, ensuring that the model prioritizes correctly identifying rare cases. Additionally, ensemble methods like boosting iteratively correct errors made on the minority class, improving overall classification performance. Properly handling missing data, applying normalization or standardization, and addressing class imbalance ensures that feature selection techniques yield more reliable and interpretable results in medical datasets.

5.4 Noise Reduction and Outlier Detection

Noise in medical data can result from measurement errors or variations in clinical procedures. Techniques like smoothing (moving averages, Gaussian filters) and PCA help reduce noise while preserving key patterns. Outliers, which can distort analysis, are detected using Z-score ($|Z| > 3$) and IQR($Q1 - 1.5 \times \text{IQR},\ Q3 + 1.5 \times \text{IQR}$). Machine learning methods like isolation forests and autoencoders also identify anomalies. Reducing noise and handling outliers ensures reliable datasets, leading to more accurate and interpretable feature selection.

6 Feature Selection Techniques

6.1 Filter-Based Feature Selection Techniques

Filter-based feature selection techniques evaluate the relevance of features independently of any machine learning algorithm, using statistical measures like correlation, chi-square, or mutual information. These methods are computationally efficient and suitable for high-dimensional datasets.

6.2 Statistical Methods

Statistical methods assess the relationship between features and target variables using hypothesis testing and information theory.

a. ***Chi-Square Test***: This method assesses the independence between categorical features and the target variable, aiding in the selection of features that exhibit a strong association with the output class. It is commonly used in medical datasets with discrete variables, such as disease presence or absence.
b. ***ANOVA (Analysis of Variance)***: ANOVA is used to compare the means of a numerical feature across multiple groups of a categorical target variable. It helps in determining whether the variation in the feature values significantly contributes to classification. This method is beneficial in medical studies where feature distributions vary across disease categories.
c. ***Mutual Information***: Mutual information quantifies the dependency between a feature and the target variable, measuring the amount of information a feature provides about the target outcome. This makes it an effective method for selecting the most informative attributes in medical datasets.

6.3 Correlation-Based Methods

Correlation-based methods examine the strength and direction of the relationships between features and target variables.

a. ***Pearson Correlation***: This method measures the linear relationship between numerical features and the target variable. High absolute correlation values indicate strong associations, helping to retain the most relevant features and discard redundant ones.
b. ***Spearman Rank Correlation***: Unlike Pearson correlation, Spearman correlation assesses monotonic relationships, making it suitable for nonlinear associations often found

in medical data. It is particularly useful in ranking features based on their predictive importance.

6.4 Information Gain and Entropy-Based Methods

Information gain is derived from entropy and measures the reduction in uncertainty when using a specific feature for classification. Features with higher information gain contribute more to decision-making and are preferred for model training. These methods are widely used in decision trees and random forests to rank feature importance in medical applications, such as disease classification.

6.5 Application of Filter Methods in Medical Data

Filter methods are extensively applied in medical datasets due to their efficiency and scalability. They are commonly used for biomarker identification, gene expression analysis, and clinical feature selection. By removing irrelevant and redundant features, these techniques improve model interpretability and reduce computational costs, making them highly valuable for predictive healthcare models.

6.6 Wrapper-Based Feature Selection Techniques

Wrapper-based feature selection techniques evaluate feature subsets by training a model and assessing its performance. Though more computationally expensive, wrapper methods can capture complex interactions between features, making them effective for improving model accuracy, but less efficient in terms of computation compared to filter methods.

6.7 Overview of Wrapper Methods

Wrapper methods use a predictive model to evaluate different feature subsets iteratively. They search for the optimal feature combination by training models multiple times, making them computationally intensive but highly effective in capturing feature interactions. These methods often outperform filter methods in predictive accuracy but require significant computational resources.

6.8 Algorithm of Wrapper Methods

(a) ***Recursive Feature Elimination (RFE)***: RFE ranks features based on their importance and removes the least significant ones iteratively. It is widely used in medical applications where feature selection impacts model generalization, such as MRI-based disease diagnosis (Taloba and Matoog 2025).
(b) ***Sequential Feature Selection (Forward/Backward Selection)***: Forward selection begins with an empty feature set and adds features incrementally, selecting those that enhance model performance. Backward elimination starts with all features and removes the least significant ones. These methods are useful when the feature space is not too large, as they help refine medical datasets without excessive computation.
(c) ***Genetic Algorithms (GA)***: GA mimics the process of natural selection to identify the optimal subset of features. It evolves feature sets through selection, crossover, and mutation, ensuring optimal selection for complex medical data. GA is often applied in genomic studies and personalized medicine.
(d) ***Particle Swarm Optimization (PSO)***: PSO is inspired by swarm intelligence and optimizes feature selection by iteratively improving candidate solutions. It is effective in high-dimensional datasets, such as medical imaging and multi-omics data analysis.

6.9 Pros and Cons of Wrapper Methods in Medical Data

Wrapper methods offer higher accuracy by capturing feature dependencies, but are computationally expensive. They are prone to overfitting when dealing with small datasets, making them less suitable for certain medical applications. However, when applied to well-curated datasets, these methods can significantly enhance predictive performance in disease classification and risk assessment.

7 Hybrid and Advanced Feature Selection Techniques

Hybrid feature selection techniques integrate filter, wrapper, and embedded methods to leverage their individual strengths while mitigating their limitations. Filters provide an initial

ranking of features based on statistical relevance, wrappers refine the selection through iterative model evaluation, and embedded methods ensure selection during training. This combination enhances performance, particularly in medical applications where datasets are high-dimensional and heterogeneous. Hybrid approaches are commonly used in predictive modeling for disease diagnosis, where both statistical relevance and predictive power are crucial (https://www.sciencedirect.com/science/article/abs/pii/S1532046425000292).

7.1 Deep Learning-Based Feature Selection

(a) ***Auto Encoders***: Auto encoders are neural networks that learn a compressed representation of data through unsupervised learning. By training an encoder-decoder architecture, they identify the most informative features while reducing redundancy. In medical imaging, autoencoders extract key patterns from MRI and CT scans, helping in disease classification.
(b) ***Attention Mechanisms***: Attention-based models assign different weights to features based on their importance. These mechanisms are widely used in deep learning models for genomics, where they help prioritize critical genetic variations related to disease progression. Attention mechanisms in medical NLP applications also enhance feature selection by concentrating on the most relevant features and clinical terms in patient records.

7.2 Multi-objective Optimization for Feature Selection

Multi-objective optimization methods aim to balance competing goals, such as maximizing predictive accuracy while minimizing the number of selected features. Techniques like Pareto-based optimization and evolutionary algorithms, including NSGA-II, are used to find optimal feature subsets. These methods are particularly beneficial in medical research, where feature selection must balance interpretability, computational efficiency, and clinical relevance.

7.3 Applications in Multi-modal Medical Data

Multi-modal medical data, such as integrating genomic, imaging, and clinical data, requires advanced feature selection techniques to extract meaningful insights. Hybrid and deep learning-based methods are used to fuse information from different sources while selecting the most informative features. Applications include personalized medicine, where genomic data is combined with MRI features for cancer diagnosis, and multi-modal disease prediction models that integrate electronic health records with imaging data. These approaches improve diagnostic accuracy and enable more precise treatment strategies.

7.4 Embedded Feature Selection Techniques

Embedded feature selection techniques incorporate feature selection into the model training process, enabling simultaneous learning and ranking of features. Unlike filter and wrapper methods, embedded techniques leverage model-specific properties to identify the most important features, improving both efficiency and predictive accuracy. These methods are particularly useful for handling high-dimensional medical datasets, such as genomic and imaging data.

7.5 Regularization Techniques

Regularization methods penalize model complexity by adding constraints to the learning process, reducing overfitting and improving feature selection.

(a) ***Lasso (L1 Regularization)***: Lasso regression applies L1 regularization, which drives some feature coefficients to zero, effectively eliminating irrelevant features. This method is widely used in medical datasets where feature sparsity is desirable, such as selecting key biomarkers in genetic studies (Ghosh et al. 2025).
(b) ***Ridge (L2 Regularization)***: Ridge regression employs L2 regularization, which penalizes large feature weights without eliminating them. While it does not perform feature selection directly, it helps stabilize models when features are highly correlated, making it valuable for complex medical predictions (Zhao and Li 2024).
(c) ***Elastic Net***: Elastic Net combines L1 and L2 regularization to balance feature selection and stability. It is particularly useful in medical applications where datasets have multicollinearity or when the number of features exceeds the number of samples, such as in high-dimensional gene expression analysis (Wu et al. 2025).

7.6 Tree-Based Methods

During training, tree-based models automatically pick features by balancing each feature's importance.

Table 2 Summary of feature selection techniques for different medical data types

Medical data type	Common feature selection techniques	Description	Algorithms used
Genomic and proteomic data (Bohrer and Dorn 2024; Islam et al. 2024)	LASSO (L1 regularization), mutual information, chi-square test	These methods are used to identify significant genes or proteins that contribute to disease classification. LASSO reduces the feature space by shrinking coefficients	LASSO, scikit-learn, R (limma package)
Medical imaging (MRI, CT, X-ray) (Danaci et al. 2025)	PCA (principal component analysis), deep learning-based methods (CNNs), random forest	PCA is used for dimensionality reduction, while CNNs are often used to automatically extract important features from images for classification tasks	TensorFlow, PyTorch, OpenCV, scikit-learn
Electronic health records (EHR) (Allam et al. 2024)	Random forest, XGBoost, gradient boosting machines (GBM)	Tree-based methods such as Random Forest and GBM are used for identifying important clinical features that contribute to disease prediction or risk stratification	Scikit-learn, XGBoost, LightGBM
Time-series data (ECG, EEG) (Ji and Zhu 2024)	Recursive feature elimination (RFE), long short-term memory (LSTM), autoregressive models	Time-series models like LSTM can extract temporal dependencies in data. RFE can be used for removing less relevant features in recursive steps	Keras, TensorFlow, scikit-learn

(a) ***Random Forest***: Random Forest consists of multiple decision trees and ranks feature importance based on how often features are used in decision splits across all the trees. This technique is highly effective in medical diagnostics, where different clinical factors contribute to disease prediction.

(b) ***Gradient Boosting Machines (GBM)***: GBM builds decision trees sequentially, focusing on reducing errors in each iteration. It assigns importance scores to features based on their contribution to enhancing model performance. GBM-based feature selection is widely used in radionics and personalized medicine.

7.7 Feature Selection for Specific Medical Data Types

Feature selection methods need to be tailored to the specific characteristics of the medical data at hand. In genomic and proteomic data, high dimensionality and noise present challenges, requiring methods like mutual information and regularization techniques to identify relevant biological features. In medical imaging, such as MRI, CT, and X-rays, feature selection focuses on extracting key patterns that contribute to diagnosis or segmentation, using methods like convolutional neural networks (CNNs) or deep learning-based feature selection. For electronic health records, feature selection must handle complex, structured data with categorical, numerical, and text-based information, often leveraging ensemble methods or tree-based models. Time-series data (Yasmin et al. 2023) like ECG and EEG require specialized approaches to capture temporal dependencies, often using techniques like recursive feature elimination (RFE) or recurrent neural networks (RNNs) for feature extraction. Table 2 presented the summary of different feature selection techniques, and Table 3 explains the challenges in Feature Selection for Medical Data and Potential Solutions.

8 Conclusion and Summary

Feature selection is vital in medical data analysis, improving model accuracy, reducing complexity, and enhancing interpretability. This book explored filter-based, wrapper-based, embedded, and hybrid techniques, highlighting their applications in genomics, medical imaging, electronic health records, and time-series data like ECG and EEG. Proper feature selection aids disease diagnosis, prognosis, and treatment recommendations while addressing challenges like class imbalance and high dimensionality. Selecting the right method depends on the dataset. LASSO, mutual information, and entropy-based methods work well for genomic data, while PCA and CNNs excel in medical imaging. Decision trees and gradient boosting benefit EHR analysis, whereas auto encoders and LSTMs are effective for time-series data. Ensuring data preprocessing, validating features with domain experts, and using cross-validation improve selection reliability. Despite progress, challenges persist. Scalability, integration of multimodal data, and handling noisy or missing values require advanced techniques. Ethical concerns, particularly in federated learning, demand privacy-preserving feature selection. The future of feature selection in medical AI lies in combining

Table 3 Challenges in feature selection for medical data and potential solutions

Challenge	Description	Potential solutions
Overfitting (Abdel-Aal 2005)	Models may become too complex, leading to poor generalization on new data	Regularization techniques such as lasso and ridge, use of cross-validation to evaluate model robustness
Scalability (Sun and Wu 2023)	Medical datasets are often large and high-dimensional, making feature selection computationally expensive	Use of dimensionality reduction methods (e.g., PCA) and distributed computing frameworks like Apache Spark
Missing data (Hasan et al. 2025)	Medical data often contains missing or incomplete records	Imputation techniques like KNN imputation or using algorithms that handle missing values, such as random forest
Interpretability (https://www.sciencedirect.com/science/article/pii/S0306261924006974)	Some feature selection methods (e.g., deep learning) result in black-box models that are hard to interpret	Use of explainable AI techniques like SHAP or LIME to interpret model decisions and feature importance

traditional methods with deep learning and explainable AI to enhance disease prediction, precision medicine, and clinical decision-making.

References

Abdel-Aal RE (2005) GMDH-based feature ranking and selection for improved classification of medical data. ScienceDirect 38(6):456–468. https://doi.org/10.1016/j.jbi.2005.03.003

Akter SB et al (2025) Fair and explainable myocardial infarction (MI) prediction: novel strategies for feature selection and class imbalance correction. Comput Biol Med 184:109413. https://doi.org/10.1016/j.compbiomed.2024.109413

Allam AH, Gomaa I, Zayed HH, Taha M (2024) IoT-based eHealth using blockchain technology: a survey. Cluster Comput 27(6):7083–7110. https://doi.org/10.1007/s10586-024-04357-y

Bohrer JdS, Dorn M (2024) Enhancing classification with hybrid feature selection: a multi-objective genetic algorithm for high-dimensional data. Expert Syst Appl 255:124518. https://doi.org/10.1016/j.eswa.2024.124518

Danaci C, Avci D, Tuncer SA (2025) Diagnosis of pes planus from X-ray images: enhanced feature selection with deep learning and machine learning techniques. Biomed Signal Process Control 106:107769. https://doi.org/10.1016/j.bspc.2025.107769

Ergul Aydin Z, Kamisli Ozturk Z (2024) Filter-based feature selection methods in the presence of missing data for medical prediction models. Multimed Tools Appl 83(8):24187–24216. https://doi.org/10.1007/s11042-023-15917-6

Ghosh N, Kumar Mridha S, Paul R (2025) LASSO-mCGA: machine learning and modified compact genetic algorithm-based biomarker selection for breast cancer subtype classification. IEEE Access 13:17673–17682. https://doi.org/10.1109/ACCESS.2025.3532361

Hasan M, Yasmin F, Deng L (2023) Rule mining of early diabetes symptom and applied supervised machine learning and cross validation approaches based on the most important features to predict early-stage diabetes. IJIRMPS - Int J Innov Res Eng Multidiscip Phys Sci 11(3). https://doi.org/10.37082/IJIRMPS.v11.i3.230225

Hasan M et al (2025) Enhancing stroke disease classification through machine learning models via a novel voting system by feature selection techniques. PLoS ONE 20(1). https://doi.org/10.1371/journal.pone.0312914

Hema K, Meena K, Pandian R (2024) Analyze the impact of feature selection techniques in the early prediction of CKD. Int J Cogn Comput Eng 5:66–77. https://doi.org/10.1016/j.ijcce.2023.12.002

Hossain MI, Zamzmi G, Mouton PR, Salekin MS, Sun Y, Goldgof D (2025) Explainable AI for medical data: current methods, limitations, and future directions. ACM Comput Surv 57(6)148:1–148:46. https://doi.org/10.1145/3637487

Innovative framework for accurate and transparent forecasting of energy consumption: a fusion of feature selection and interpretable machine learning. ScienceDirect. https://www.sciencedirect.com/science/article/pii/S0306261924006974

Islam MdA, Majumder MdZH, Miah MdS, Jannaty S (2024) Precision healthcare: a deep dive into machine learning algorithms and feature selection strategies for accurate heart disease prediction. Comput Biol Med 176:108432. https://doi.org/10.1016/j.compbiomed.2024.108432

Ji W, Zhu D (2024) ECG classification exercise health analysis algorithm based on GRU and convolutional neural network. IEEE Access 12:59842–59850. https://doi.org/10.1109/ACCESS.2024.3392965

Karampour RA, Bakhtiar AA, Hajizamani K, Fallahi A, Zadeh HS (2024) Cardiac ischemia detection using self-organizing map feature selection method. In: in 2024 31st national and 9th international Iranian conference on biomedical engineering (ICBME), pp 57–62. https://doi.org/10.1109/ICBME64381.2024.10895018

Khan S-R, Khan Z, Hossain MZ, Mayumu N, Yasmin F, Aziz Y (2024) Boosting the prediction of brain tumor using two stage BiGait architecture. In: 2024 International conference on digital image computing: techniques and applications (DICTA), pp 411–418. https://doi.org/10.1109/DICTA63115.2024.00067

Kumar Sahu P, Fatma T (2025) Optimized breast cancer classification using PCA-LASSO feature selection and ensemble learning strategies with optuna optimization. IEEE Access 13:35645–35661. https://doi.org/10.1109/ACCESS.2025.3539746

Natarajan K, Baskaran D, Kamalanathan S (2025) An adaptive ensemble feature selection technique for model-agnostic diabetes prediction. Sci Rep 15(1):6907. https://doi.org/10.1038/s41598-025-91282-8

Nemati K, Sheikhani AHR, Kordrostami S, Roudposhti KK (2024) The embedded feature selection method using ANT colony optimization with structured sparsity norms. Computing 107(1):29. https://doi.org/10.1007/s00607-024-01387-7

Pak A, Rad AK, Nematollahi MJ, Mahmoudi M (2025) Application of the Lasso regularisation technique in mitigating overfitting in air quality prediction models. Sci Rep 15(1):547. https://doi.org/10.1038/s41598-024-84342-y

Petmezas G et al. Enhanced heart failure mortality prediction through model-independent hybrid feature selection and explainable machine learning. https://www.sciencedirect.com/science/article/abs/pii/S1532046425000292

Rabiee M, Mirhashemi M, Pangburn MS, Piri S, Delen D (2024) Towards explainable artificial intelligence through expert-augmented supervised feature selection. Decis Support Syst 181:114214. https://doi.org/10.1016/j.dss.2024.114214

Rahman A, Rahman MM, Yasmin F (2024) Revolutionizing healthcare through federated learning: a secure and collaborative approach. In: Federated deep learning for healthcare. CRC Press

Rahmat F et al (2024) Supervised feature selection using principal component analysis. Knowl Inf Syst 66(3):1955–1995. https://doi.org/10.1007/s10115-023-01993-5

Salsabila B, Said N, Abdelali Z (2025) Optimizing breast cancer diagnosis: harnessing the power of nature-inspired metaheuristics for feature selection with soft voting classifiers. Int J Cogn Comput Eng 6:1–20. https://doi.org/10.1016/j.ijcce.2024.09.005

Singh LK, Khanna M, Singh R (2024) Efficient feature selection for breast cancer classification using soft computing approach: a novel clinical decision support system. Multimed Tools Appl 83(14):43223–43276. https://doi.org/10.1007/s11042-023-17044-8

Song Y-W, Wang J-S, Qi Y-L, Wang Y-C, Song H-M, Shang-Guan Y-P (2025) Serial filter-wrapper feature selection method with elite guided mutation strategy on cancer gene expression data. Artif Intell Rev 58(4):111. https://doi.org/10.1007/s10462-024-11029-1

Sun L, Wu J (2023) A scalable and transferable federated learning system for classifying healthcare sensor data. IEEE J Biomed Health Inform 27(2):866–877. https://doi.org/10.1109/JBHI.2022.3171402

Taloba AI, Matoog RT (2025) Detecting respiratory diseases using machine learning-based pattern recognition on spirometry data. Alex Eng J 113:44–59. https://doi.org/10.1016/j.aej.2024.11.009

Wang C et al (2025) Multi-feature fusion RFE random forest for schizophrenia classification and treatment response prediction. Sci Rep 15(1):8594. https://doi.org/10.1038/s41598-025-89359-5

Wu Y, Kang F, Zhu S, Li J (2025) Data-driven deformation prediction model for super high arch dams based on a hybrid deep learning approach and feature selection. Eng Struct 325:119483. https://doi.org/10.1016/j.engstruct.2024.119483

Yasmin F et al (2023a) GastroNet: gastrointestinal polyp and abnormal feature detection and classification with deep learning approach. IEEE Access 11:97605–97624. https://doi.org/10.1109/ACCESS.2023.3312729

Yasmin F et al (2023b) PoxNet22: a fine-tuned model for the classification of monkeypox disease using transfer learning. IEEE Access 11:24053–24076. https://doi.org/10.1109/ACCESS.2023.3253868

Yasmin F et al (2023) AQIPred: a hybrid model for high precision time specific forecasting of air quality index with cluster analysis. Hum-Cent Intell Syst 3(3):275–295. https://doi.org/10.1007/s44230-023-00039-x

Zhao H, Li P (2024) An efficient class-dependent learning label approach using feature selection to improve multi-label classification algorithms. Cluster Comput 28(1):38. https://doi.org/10.1007/s10586-024-04756-1